

Acceptance Tests – WBOCRM

This document maps READ acceptance criteria to implemented test coverage.
All tests are located in `backend/tests/` and results saved in `backend/test_results/`.

Test Suite Summary

Suite	File	Tests	Status
Unit	tests/unit/authService.test.js	6	✓ PASS
Unit	tests/unit/scoringService.test.js	6	✓ PASS
Unit	tests/unit/ticketService.test.js	6	✓ PASS
Functional	tests/functional/api.test.js	6	✓ PASS
BDD	tests/features/auth_login.feature	4 scenarios	✓ PASS
BDD	tests/features/lead_registration.feature	2 scenarios	✓ PASS
BDD	tests/features/lead_scoring.feature	3 scenarios	✓ PASS
BDD	tests/features/ticket_creation.feature	2 scenarios	✓ PASS
Total		35 tests / 11 BDD scenarios	✓ All pass

BDD Acceptance Scenarios

Feature: Lead Registration (`lead_registration.feature`)

Scenario 1: Successfully register a new lead

```
Given I am authenticated as a sales user
When I submit a new lead with email "bdd_lead@test.wbocrm" and name "BDD Lead"
Then the response status should be 201
And the response should contain a lead_id
```

Covers: US-03, FR-LM-01

Scenario 2: Reject duplicate email registration

```
Given a lead with email "bdd_dup@test.wbocrm" already exists
When I submit a new lead with email "bdd_dup@test.wbocrm" and name "Dup Lead"
Then the response status should be 400
And the response error should be "DUPLICATE_EMAIL"
```

Covers: US-03, FR-LM-02

Feature: Support Ticket Creation (`ticket_creation.feature`)

Scenario 1: Successfully create a support ticket

Given I am authenticated as a support user
And a lead exists in the system
When I create a ticket with description "BDD test ticket" and priority "Low"
Then the response status should be 201
And the response should contain a ticket_id

Covers: US-11, FR-ST-01

Scenario 2: Update a ticket status

Given an existing support ticket
When I update the ticket status to "In Progress"
Then the response status should be 200
And the ticket status should be "In Progress"

Covers: US-12, FR-ST-02

Feature: User Authentication (auth_login.feature)

Scenario 1: Successful login with valid credentials

When I log in with email "bdd_auth_sales@test.wbocrm" and password "BddAuth123!"
Then the response status should be 200
And the response should contain a token
And the token payload should include role "sales"

Covers: US-01, FR-AUTH-01

Scenario 2: Failed login with invalid password

When I log in with email "bdd_auth_sales@test.wbocrm" and password "WrongPass999!"
Then the response status should be 401

Covers: US-01, FR-AUTH-02

Scenario 3: Rejected request without token

When I request the leads list without a token
Then the response status should be 401

Covers: US-01, FR-AUTH-03

Scenario 4: Rejected request with insufficient role

Given I am authenticated as a support user
When I request the leads list
Then the response status should be 403

Covers: US-02, FR-AUTH-04

Feature: Lead Priority Scoring (`lead_scoring.feature`)

Scenario 1: High engagement metrics receive high score

When I create a lead with metrics calls 20 meetings 5 budget "high" companySize "enterprise" emailOpens 20
Then the response status should be 201
And the lead priority_score should be greater than 70

Covers: US-04, FR-SC-01

Scenario 2: No engagement receives low score

When I create a lead with metrics calls 0 meetings 0 budget "low" companySize "small" emailOpens 0
Then the response status should be 201
And the lead priority_score should be less than 30

Covers: US-04, FR-SC-01

Scenario 3: Leads list sorted by priority score descending

When I retrieve all leads
Then the response status should be 200
And leads should be ordered by priority_score descending

Covers: US-04, FR-LM-03, FR-SC-01

Functional Test Coverage (`api.test.js`)

Test	Expected	Covers
POST /api/leads – valid lead	201 + lead_id	US-03
POST /api/leads – duplicate email	400 DUPLICATE_EMAIL	US-03
GET /api/leads – sorted DESC	200 + ordered array	US-04
POST + PUT /api/tickets/:id	201 then 200 + status updated	US-11, US-12
GET /api/leads – no token	401	US-01
GET /api/leads – wrong role	403	US-02

Unit Test Coverage

authService (6 tests)

- `hashPassword` returns a bcrypt hash

- `comparePasswords` returns true for matching password
- `comparePasswords` returns false for wrong password
- `generateToken` returns a signed JWT
- `generateToken` payload contains `user_id` and `rbac_role`
- `verifyToken` returns the original payload

scoringService (6 tests)

- Returns 0 for empty metrics
- Caps call score at 25 pts
- Caps meeting score at 25 pts
- Applies correct company size scores (small/medium/enterprise)
- Returns maximum 100 for fully maxed metrics
- Rounds to integer

ticketService (6 tests)

- Creates a ticket successfully
- Returns `TICKET_NOT_FOUND` for missing ticket
- Returns `CONFLICT` when `updated_at` does not match
- Updates status successfully on valid `updated_at`
- Validates status enum values
- Assigns ticket to user correctly

Non-Functional Requirements Verified

NFR ID	Requirement	Verification
NFR-ST-05	Optimistic locking on ticket updates	Unit test: <code>CONFLICT</code> on stale <code>updated_at</code>
NFR-ST-12	Dashboard auto-refresh every 2s	<code>DashboardPage setInterval(2000)</code> implementation
NFR-SEC-01	Passwords hashed with <code>bcrypt</code>	Unit test: <code>hashPassword</code> returns non-plaintext
NFR-SEC-02	JWT auth on all protected routes	Functional test: 401 without token
NFR-SEC-03	Role enforcement	Functional test: 403 for insufficient role